

32180 ORION



Connect



Reach



Sustain



Innovate



Control



Design



Meet the Team



As shown from left to right:

Ben Sparks, Abbey Stettler, Garrett Tosh, Sage Ryker, Dravyn Hermeena, Nathan Couch, Mike Brandt, Alexander Tichindelean, Anthony Nguyen

Students Not Pictured:

Ryker Aeschlimann, Caroline Kerr, Olivia Shifflett, Joshua Theophilus John

Subteam Structure:

Members apply to be on Orion in April-May. Students are asked to apply for two specific subteams. In the case where a subteam "fills up", an individual can appeal the decision by applying for a different subteam.

Members are required to select a primary and secondary subteam.

3 Subteams:

- Business
- Mechanical
- Programming

This promotes cross-training throughout the team so all team members have participated in each portion of the team. Cross-training is an important part of Orion as it builds a strong foundation of skills within team members.

Business:

- Pages: 3, 5-7

Mechanical:

- Pages: 8 - 11

Programming:

- Pages: 12

Leadership

STEP

01

Develop deep knowledge and technical competence within respective subteam by actively learning its systems, tools, and workflows. This includes: practicing core skills, taking initiative, and seeking mentorship.

STEP

02

Pursue a formal leadership role that aligns with demonstrated skills, commitment, and areas of expertise and complete the application by the due date (in July/August for most positions).

STEP

03

Engage in a formal interview process (if necessary) following the official application, where candidates can articulate their leadership philosophy, experience, and approach to problems. This interview is conducted by a predetermined committee of mentors and leaders.

STEP

04

Accept the leadership role with humility, gratitude, and a commitment to serve the team rather than lead for status or awards. Approach the position with responsibility by setting clear goals and maintaining high standards for both personal success and the success of the team.



Garrett Tosh
Mechanical Lead
'27



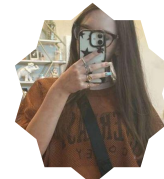
Sage Ryker
Team Lead
'27



Alex Tichindelean
Programming Lead
'27



Joshua Theophilus John
Integration Director '26

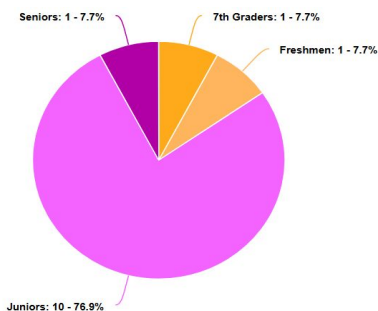


Caroline Kerr
Awards Director &
Historian '29

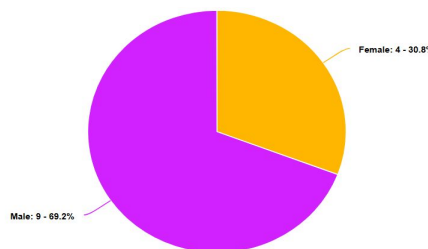


Abbey Stettler
Drive Coach '27

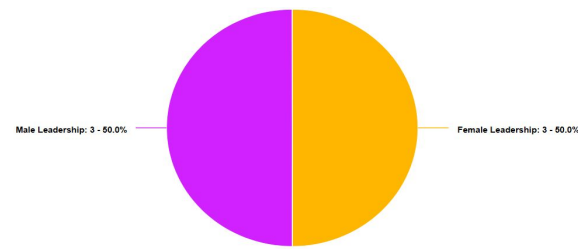
Grade Level Distribution



Gender Distribution



Gender:Leadership Ratio



Mission Statement

Orion | NorthStarRobotics (FTC 32180)'s primary goal is to inspire and empower students throughout the community by engaging them in all aspects of robotics, with a strong emphasis on STEAM: Science, Technology, Engineering, Art, and Mathematics.

Our organization (NorthStarRobotics) began with a single FIRST Tech Challenge (FTC) team and is dedicated to growing into a multi-team alliance throughout Northern Indiana, expanding opportunities for students to explore innovation, collaboration, and leadership via hands-on robotics experiences!



Sustainability

Recruiting

Recruitment for Orion | NorthStarRobotics follows a closely outlined application process:

- Early Recruitment Application (April-June)
- Standard Recruitment Application (June-July)
- Late Recruitment Application (August-September)

Members are recruited via:

- Word-of-mouth strategies
- Social Media outreach
- Established connections with two local Homeschool Co-Ops
 - Northpoint Christian Academy
 - The Wisdom Builders

01

Fundraising

Each Student is responsible for \$400 for individual Student Dues which can be covered in the following ways:

- Individual Business Sponsorships
- Fundraising Events
- Donations made in Students Name

This year we reached out to 15 companies in August and September in which received a one sponsorship of \$500. Six of these companies that did not sponsor stated that they would sponsor in the Spring.

We received a total of \$1,755 in donations and \$2,025 from two grants. The FTC SIM Argosy Rookie Team Grant also served to teach our Programming Subteam skills for build season.

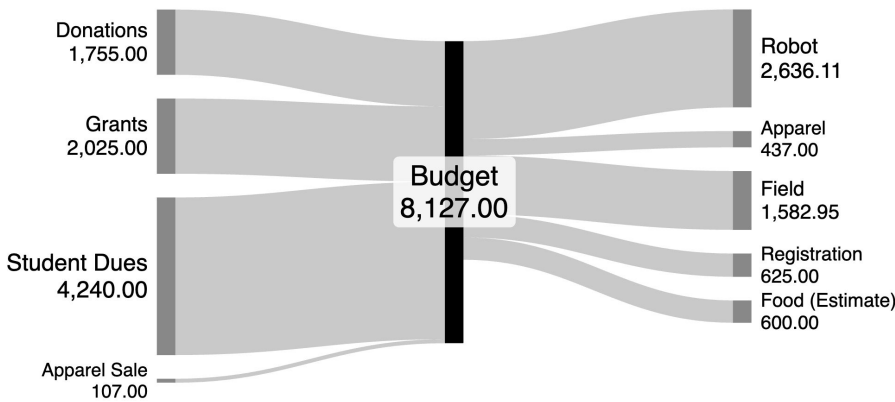
02

Teaching Initiatives

Our leadership team is committed to delivering comprehensive training across all areas of CADing (Fusion 360), Manufacturing/Assembly, Design Process, Programming (Java), Website Design (Markdown), while also emphasizing the development of essential business, marketing, and fundraising skills.

This holistic approach ensures that every team member gains both strong technical foundations and the organizational knowledge needed to sustain and grow the team. Each member of the leadership team is responsible for extensively documenting all work completed throughout the season to ensure that all knowledge is maintained and built upon.

03



Amount	Date	Account	Vender	Amount	Date	Account	Vender
\$ 400.00	9/2/2025	FORUM	Sage Do	\$ (1,064.00)	10/22/2025	FIRST	AndyMai
\$ 120.00	9/2/2025	FORUM	Garrett T	\$ 250.00	10/25/2025	FORUM	Anonym
\$ 400.00	9/2/2025	FORUM	Ben Spa	\$ (585.76)	11/3/2025	FORUM	GoBilda
\$ 400.00	9/2/2025	FORUM	Anthony	\$ (518.95)	11/4/2025	FORUM	AndyMai
\$ 120.00	9/2/2025	FORUM	Nathan C	\$ 1,000.00	11/6/2025	HackC...	Argosy F
\$ 120.00	9/2/2025	FORUM	Olivia Sh	\$ (300.00)	11/19/2025	FIRST	FIN
\$ 400.00	9/2/2025	FORUM	Alex Tich	\$ 100.00	11/21/2025	FORUM	Olivia Sh
\$ 120.00	9/2/2025	FORUM	Joshua T	\$ (195.60)	11/24/2025	HackC...	AndyMai
\$ 1,500.00	9/11/2025	FIRST	Anonym	\$ (315.32)	12/1/2025	HackC...	REV Rot
\$ 1,025.00	9/15/2025	FIRST	Castle F	\$ 0.10	11/30/2025	FORUM	FORUM
\$ 120.00	9/17/2025	FORUM		\$ (676.61)	1/2/2026	FORUM	GoBilda
\$ 100.00	9/22/2025	FORUM	Olivia Sh	\$ (34.36)	1/17/2026	FORUM	GoBilda
\$ 400.00	9/22/2025	FORUM	Caroline	\$ (8.01)	1/21/2026	FORUM	Amazon
\$ 280.00	9/22/2025	FORUM	Nathan C	\$ (138.87)	1/22/2026	FORUM	GoBilda
\$ 280.00	9/22/2025	FORUM	Garrett T	\$ 200.00	1/28/2026	FORUM	Abbey S
\$ 400.00	9/22/2025	FORUM	Ryker Ae	\$ (437.00)	1/30/2026	HackC...	Connect
\$ (325.00)	9/24/2025	FIRST	FIRST	\$ 107.00		FORUM	Families
\$ (681.58)	9/30/2025	FIRST	FIRST	\$ (600.00)		FORUM	MISC
\$ 5.63	9/30/2025	FIRST	Bryce Ca				
\$ 200.00	10/2/2025	FORUM	Joshua T				
\$ 280.00	10/2/2025	FORUM	Mike Bra				

*Amounts with parentheses are expenses

Risk Analysis

Risk Area	Concern (1-3)	Impact (1-3)	Score	Risk Level
Space	3	3	9	Critical
Funding	2	3	6	High
Students	2	2	4	Moderate
Mentors	2	2	4	Moderate
Student Interest	2	3	6	High
Knowledge Loss	1	3	3	Low

Risk Mitigation & Solutions

Rookie Team

Due to being a rookie team, it was very difficult to find sponsors as we had very little proof to back us up as a "worth-it" investment. Thus, we turned to grants to help fund us for this year. Furthermore, this will not remain a problem for next year as we will have experience to send to sponsors.

Space

This year, we did not have a specific location that we could all meet outside of members houses for several hours. While we have not found a solution to this yet, we have narrowed down a list of nearby locations that we will reach out to request location. In the meantime, we have gained access to our local Lions Club for one hour a week.

Time

Due to our location problem, we had very limited time to manufacture, program, and test our robot before our first competition. In order to combat this, we utilized schedules and the GitHub Task Board to keep us on track and get our robot complete before our deadline. We also implemented Virtual full team meetings to maintain communication.

Knowledge

As a rookie team, none of our team members had knowledge within FTC. o combat this each of our team members researched a different part of FTC to ensure that we could compete well against others. As part of this, we created connections with other FTC teams in order to learn how their practices could better ours.

Team Goals

Summer (Pre)Season: Planning for the upcoming seasons, Member registration, Leadership selection, Subteam education.

Practice Season: Drive team selection, Strategy team selection, Driver/Human Player practice.

May - August

November - January

September - November

January - April

Build Season: Kickoff, Brainstorming, CADing, Manufacturing, Programming, Testing.

Competition Season: Qualifying Competitions, FIN State Championship, FIRST World Championship

Robot Goals

- Achieve a smoother shooter
- Achieve a consistent 6 piece autonomous
- Refine the Robot design by utilizing match and test results
- Perfect Odometry system
- Develop a system to know what wire connects where

Outreach Goals

Volunteer Engagement

This year we had **31% of our students volunteer** for events across FIRST, Lions Club events, and the ISA.

While we have been able to **reach countless FIRST teams** and individuals across our community (with an estimate around: 30 FLLC teams, 31 FTC teams, 23 FRC teams, ~209 other members of our community that are not FIRST based)! We would love more involvement across the team in volunteering and our goal is to boost ourselves to 62% volunteer engagement next year.

Youth in STEAM

Part of our mission as a team is to **improve Youth in STEAM engagement** and knowledge. It is incredibly important to us that we ensure that every member of our community is able to explore their passions.

Thus, we have made it a goal to hold Youth in STEAM workshops in order to get more individuals involved. Furthermore, we have made it a mission to found a 501(c)(3) that will help stand up other Robotics teams, whether FLLC, FLLC, or FTC.

FIRST Mentorship

This year we made so many connections with other FIRST teams and we have learned the value of mentorship from other teams via the **GMP (Global Mentorship Program)**. Now, we are committed to mentoring FLLC teams in our area and offering feedback to other FTC teams, such as #30458 the MasterMinds.

We want to get involved in mentorship next year in order to give back to the community that gave so much to us.

Nonprofit Goals

01

Complete and file the paperwork for the 501(c)(3) to ensure that we have proper non-profit status!



02

Gain more corporate sponsorships, allowing us to fund more outreach and robot goals!



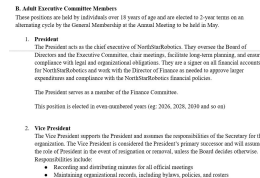
03

Stand up more FIRST teams (FTC or FLLC) and support pre-existing teams!



04

Support Parent and Alumni involvement through Board Positions and Mentorship Opportunities!



05

Continue to support our existing team and ensure that we continue to thrive for many years!



Fundraising Goals

Reach out to more businesses

- Green: Sponsored
 - Cyan: Confirmed Sponsorship for next year in prior correspondence
 - Yellow: Have indicated that we should reach out again in the future
 - Red: Declined Sponsorship
- Screenshot is not a complete list*

Business Name:	Business Name:
Modern Strength Psychology Center	GM
Goelzer Investment Management	ValGoTech
Shipley CPA Group, P.C.	Warren Williams
Resultant	Talk To Tucker
John Reno Family Dentistry	Birds
Telabotics	Clark's Concrete Pumping
Finch Automation	Sanctuary Wealth
Realize Inc.	McGinty Reitz Financial
Carter Robotics	KBIConsulting
SAAM Inc	Pence Media Group
Tosh Dental Group	
Escape Room USA	Pella
Hurco	DOZ LLC

Hold fundraising events

- Custom Merch
- Bake Sales
- Social Media Challenges
- Online Platforms

Reduce student fees to make robotics more accessible to the community. Fundraising can help us reduce fees from the current price in order to support every community member!

Expand STEAM interest and expand education opportunities.

Build relationships with local organizations, FIRST teams, and businesses.

Build technical awareness in youth interested in STEAM.

Community Goals

Volunteer throughout our community in order to ensure the FIRST message is spread.

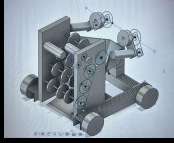
Inspire local youth to explore STEAM fields!

Team Connections

Global Mentorship Program (GMP)

The Global Mentorship Program served to connect us with two incredible international teams.

#1574 - MisCar from Israel
Mentored us: CAD, Electrical



Original CAD design from 10/22/2026

Suggestions from #1574:

Layla | MisCar 1574 22/10/2025 13:15
Hmmm. Your shooter seems structurally weak. And I'd recommend using pp dt instead of this modified steafer
And add a counter roller on intake

#19176 - Broboti from Romania
Mentored us: Robot Programming and Control system

Ioana | 19176 12/11/2025 02:27
Hi!
How's everything going? How's the season been so far? We'd love to hear how things are coming along with the robot, fundraising, and outreach, and see how we can support you to reach your goals.
Maybe we could set up a short meeting sometime soon to chat more about your progress and what you might need.
Can't wait to hear from you!



International Connections

Them: Mechanical help, fundraising help, programming & control input.

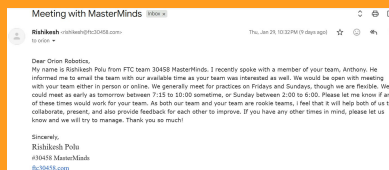
Us: Outreach help, international experience, general feedback on initiatives

Primarily connected using Discord & Email

#30458 - MasterMinds

- Local community rookie team
- Contacted us looking for help on their Portfolio and Presentation

Initial Contact



Feedback Portfolio

We offered feedback in four separate sections:

- Important notes
- Questions
- Compliments
- Low priority nitpicks

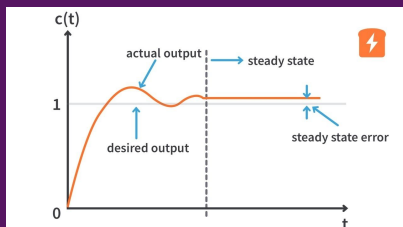
Feedback Presentation

Having not actually heard the presentation, the feedback was given to the slideshow. We offered suggestions on misspelling and capitalization errors!

Professionals: Alex Ryker

Software engineer @ Flexware Inc

Helped guide us through fixing our programming when we were having problems fixing our Autonomous and Shooter errors! His guidance helped us solve the problems without telling us what the problem was in order to fix it.



acr 04/02/2026 19:53
Okay--so to rule out some of those environmental factors, I'd recommend finding a spot of floor, marking out a straight line of a known length with a piece of painter's/masking tape, then setting up a new path in PedroPatting that does nothing but drive in a straight line for that distance. Run it on the tape line several (at least 3) times and see how it behaves. Does it drive the correct distance in the first place? If not, is it at least consistent in how much it under or overshoots? Doing that will tell us a lot about where in the system to focus our attention.

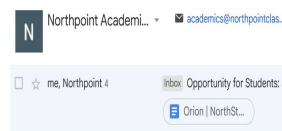
Homeschool Co-Ops:

#1

NorthPoint Christian Academy

Established an outreach connection with the NorthPoint Christian Academy and retained confirm interest for next season! After sharing our interest and goals Pamella B. ensured that they would invite students in the 2026-2027 season!

Communication

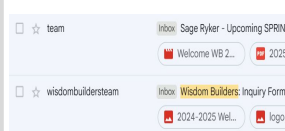


#2

The Wisdom Builders

Established an outreach connection and shared our mission of STEAM education for youth! This was an exciting connection moment for both them and us and we can't wait for future opportunities!

Communication



Mentors

Mr. Bryce Castle (Head Mentor)
Mrs. Cherie Castle
Mrs. Bryn-Hemsley Masiuk
Mr. Nick Masiuk
Mr. Adam Shifflett
Dr. Michael Tosh

Recruitment Process

Our team actively recruits mentors through strategic networking within our local community. By building genuine relationships and maintaining a strong presence with our mentors, we have created opportunities for experienced individuals to learn about our mission and get involved. Interested individuals are always more than invited to connect with us to explore mentorship roles that align with their skills and availability in order to ensure a sustainable and well-supported network.

Mentorship Structure

Director of Mentors
(Head of all mentors from the nonprofit level)
↓
Head Mentor
(Primary mentor for a specific team)
↓
Individual Mentors

Mentorship Expectations

Mentors **are** guides, **not** doers. The goal of mentorship is to help students grow their skills, confidence, and autonomy. Mentors should empower students to take initiative, make decisions, and learn from failure. Mentors should respect the student-led nature of the team rather than creating a mentor run environment. Mentors should practice active listening and ask guiding questions instead of giving direct answers when students need help in order to foster creativity and critical thinking within students.



Outreach

FIRST Events

UIndy FLLC QT

Westfield FTC QT

UIndy FLLC States

11/22/25

12/06/25

01/18/26

1 Student and 1 Mentor

2 Students and 2 Mentors

1 Student and 1 Mentor



Lion's Club Presentation

Presentation given to our local Carmel Lions Club in order to secure a partnership!



We Receive: Space for 1 hour/week

They Receive: Volunteers for various events

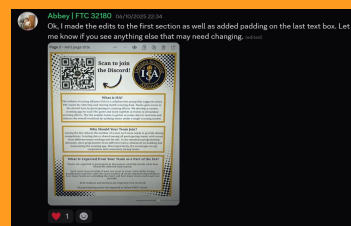
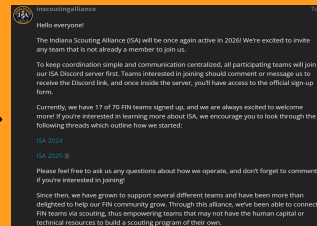
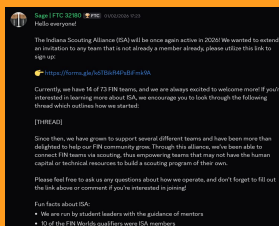
We have volunteered for:

- Bell Ringing for the Salvation Army!

We plan to volunteer for more events in the future!

Indiana Scouting Alliance

Two of our students hold leadership position in the FRC scouting alliance that 23 FRC teams from around the state are a part of! Through this, we have been able to provide a valuable service to the FRC community.



Future Plans



Continue Current Initiatives!

We want to continue and expand our volunteering efforts within FIRST events, local opportunities, and the Indiana Scouting Alliance!

Our plan is to increase our number of volunteers, and thus impact.



Youth in STEAM Events!

Potential partnerships to host Youth in STEAM Events within multiple of our local organization. Reaching out starts now, with Homeschool Co-Ops, CCPR's ESE/SCS department, and potentially hosting events at CCPL!



Mentor Local Robotics Teams (FLLC & FTC)

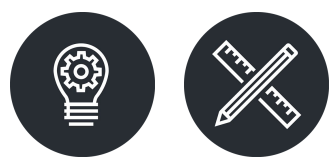
We want to get more involved in our local FIRST teams by becoming mentors for the local robotics teams. We have made connections with teams this year, and we hope to offer mentorship opportunities to them in the future!



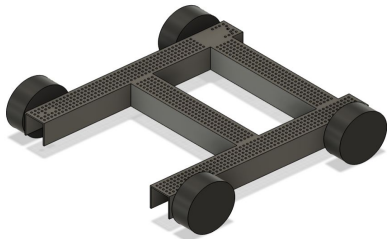
Maintain an International Presence!

We plan to switch from being a Mentee team in the GMP to a Mentor team in the GMP!

This will connect us with future rookie teams in order to help others innovate!



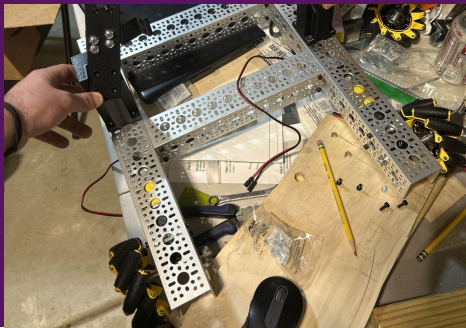
Robot Overview



Drive Train

Strategy: The intent of the chosen drive train is to have a robust and reliable base for the team to use for future seasons and games.

Compromises: The drivetrain is a GoBilda Strafer® Chassis Kit (96mm Mecanum Wheels) which is a widely used version across FTC. This limits our ability to stand out against our competitors for a reliable design that is sturdy enough to base the rest of our design off of.



Benefits	Risks
1. Smoother Auton 2. Mecanum > Tank Drive 3. Concealed motors 4. Spacious intake area 5. Protected wire lines 6. Ease of assembly 7. Robust design 8. Sturdy frame 9. Usable in future years	1. Hole pattern is a constraint 2. Does not stand out against most other robots 3. Large upfront investment 4. Uses four of the eight allotted motors

Intake Walls

Our Design Process

Brainstorm Solutions

What would make the intake sturdy?

Prototype Solutions

Found a good solution to base prototypes off of.

Test and Change

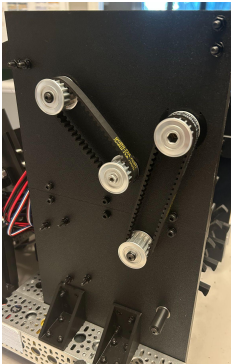
Found problems with prototypes and went through 10+ iterations.

Finalization

Finalized the Intake walls as you can see from the images.



CAD



Result

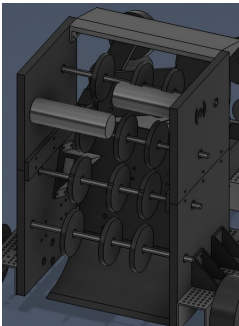
Benefits	Risks
1. Printable in-home 2. Ease of replacement 3. Learning opportunity for team 4. Placement of parts was easily adaptable 5. Strong against forces in multiple directions 6. Wall material allows for modifications to easily be made as need be 7. Good opportunity for CAD learning	1. High center of mass 2. Non reusable in future years 3. Uses a large amount of 3D printing plastic 4. Took many iterations to make final result

Wall Strategy: Have as wide of an intake as possible to reduce pressure on driver and be able to store the allotted three artifacts for use.

Wall Compromises: Chose to not use metal components to learn and push the boundaries of 3D printing and its capabilities

Stability Element: The intake walls are comprised of quarter inch 3D printed PLA to withstand excessive match forces

Intake



CAD



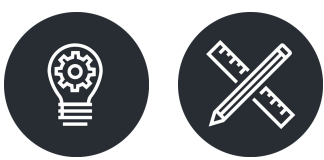
Result

Intake Strategy: Easily and reliably intake and control three artifacts with the sub strategy of being a feeder bot in the unlikely scenario of the shooter breaking.

Intake Compromises: Intake inhibits the possibility of a color sorter to simplify the design as team learns during our rookie year.

Benefits	Risks
1. Easily fits three Artifacts 2. Doubles as a surface to activate gates with 3. Durable structure protects motors 4. Motors are firmly secured	1. High center of mass 2. Artifacts rarely double up at the top jamming the intake 3. Each shaft is comprised of two smaller shafts secured by a threaded rod and 3D printed spacer

Brainstorming	Prototype	Test
Brainstorming brought up three choices for an intake; one that fed directly into the shooter, one that fed into a spindexer, and the chosen design that boasted a wide intake and plenty of room for artifacts. The design benefited from a separation of subsystems	The Intake prototype was designed around the constraints of the intake ramp at 70 degrees and the space in the base drivetrain as well as the strict 18 inch cubed restrictions. The intent was to have as wide of an intake as possible and be able to store three artifacts in the intake.	During testing the intake withstood considerable impacts but was unable to intake artifacts due to the neglect of effective tolerances. From testing many changes have been made to the intake to improve consistency and alleviate the problem of jamming.



Robot Overview

Shooter

Shooter Compromises: In this design we compromise the rate at which we can shoot artifact for a robust design and reliability that is easily implemented and proven effective of the period of a match. This robust design has proven to withstand the stresses of matches, freeing up our pit time for more driver practice and testing out new code to adapt to the need of the competition.

Benefits

1. Reliable
2. Easy to track progress
3. Robust design
4. Few moving parts
5. Directly connected to hopper and drivetrain

Risks

1. Slow fire-rate of artifacts
2. Set to a fixed angle
3. Motors are exposed
4. Takes up a large area
5. Rubbed into hopper

Design Process

Brainstorming Prototype Feedback and Testing Changes

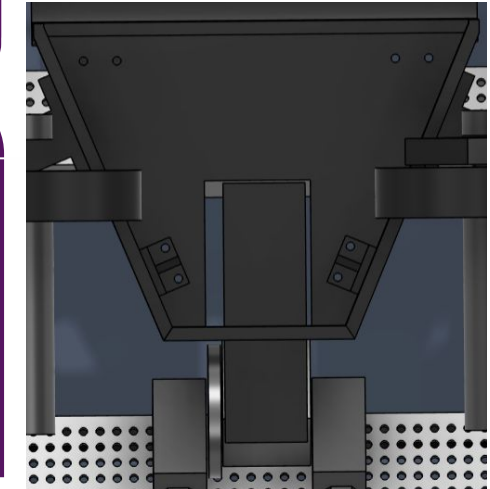
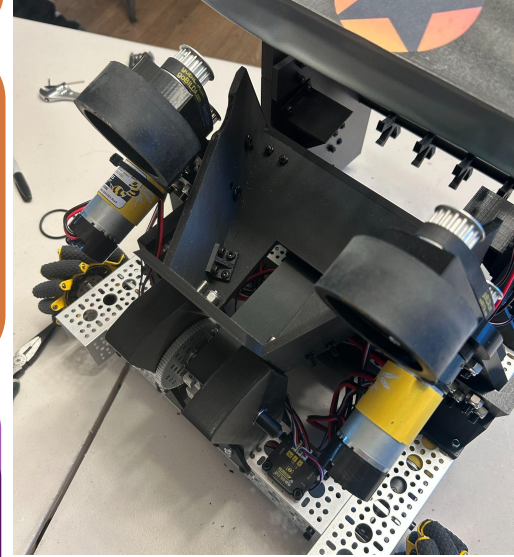
All of our brainstorming consisted of double fly wheel shooters to simplify our first year robot and reduce the number of components needed during purchasing to adhere to our tight budget.

Prototyping revealed that a fixed angle near 70 degrees took advantage of the design of field elements while also assisting in the need of a high angle for the desired intake.

Feedback from testing revealed a drastic need for a higher kinetic energy due to the artifacts being incapable of being launch more than a foot above the robot.

The shooter has not had many changes, as keystone of our design that has been proven effective with the exception of the motors needing to be replaced with ones capable of higher RPMs.

Shooter Strategy: The strategy is to reliably and consistently output a stream of artifacts. Using two shooter wheels reduces the spin on the artifacts, allowing for flight path that takes advantage of the geometry of the field element and versitiling the areas from which our robot can effectively score artifacts.



Hopper

Hopper Compromises: The hopper simplifies aspects of the robot for the compromise of slow fire-rate and no color sorting capabilities. This simplification allows the robot as a whole to be more robust and reliable.

Hopper Strategy: The hopper allows the robot add a level of redundancy to positioning the artifacts ensuring consistency in shots made by the shooter during matches. The hopper also reduces the number of moving parts in contrast to competitor robots mitigating multiple points of failure ensuring consistency and reliability during matches, competitions, and during the entire season.

Brainstorming

The hopper option appeared during brainstorming as a reliable separation from the intake and shooter. This option was appealing due to being unsure of the consistency and strain on coding to have the motors shut off when a artifact is in place to store the artifacts and not instantly eject them.

Prototyping

The hopper was effective during prototyping and allowed the team to focus on other aspects of the robot and tuning the systems.

Testing

During testing the issue of balls doubling up and jamming when all three artifacts are stored and wedges in the intake were added to address the issue.

Benefits

1. Controls position of artifacts
2. Allows for wide intake
3. Large flat surfaces allows for multiple points for attachment
4. Easy separation from intake to shooter
5. Few moving parts

Risks

1. Slow servo lever limits rate of output
2. Area of graty control instead of active control
3. Does not provide color sorting capabilities

Lessons Learned

Lesson #1

Shooter Velocity

We found that our original shooter motors were not able to consistently achieve the velocity required for accurate and reliable scoring. As a result, we upgraded to stronger motors capable of delivering higher torque and sustained speed. This change significantly improved shot consistency and overall performance during testing. With the increased velocity, we were able to meet the game's scoring demands more effectively.

Lesson #2

Prioritize CAD

As a rookie team with limited CAD experience, we initially built our robot out of wood, but scrapped the parts due to time constraints and made a final push to fully CAD the robot. During assembly, missing tolerances caused several components to fit poorly or require excessive force to secure.

Lesson #3

Dead Space

Jamming became an issue when two game pieces were taken at once and entered a dead space between the intake and hopper, where the lack of guiding walls allowed them to overlap and become stuck. To address this, we added internal guides to better direct the pieces' movement.



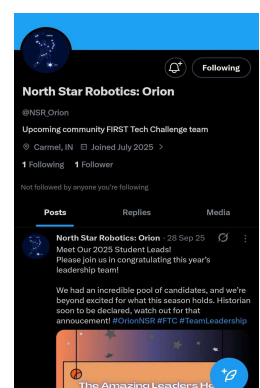
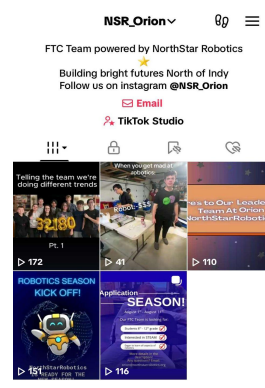
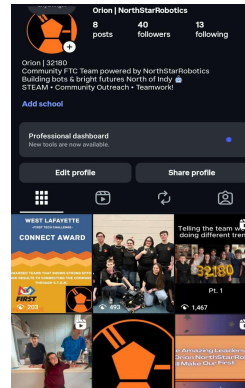
Marketing & Awards

9/15

Marketing Strategies

- 01 - Social Media engagement through unique and trendy videos. We utilize the "Content Marketing" strategy to spread awareness of FIRST and robotics.
- 02 - Email Reach-out to local businesses
- 03 - Creating Measurable Objectives that we can follow through with for Social Media, Outreach, and Volunteer engagement.
- 04 - Newsletter Updates which are sent to students, parents, mentors, and our individual partners!

Social Media: @NSR_Orion



Awards Process & Selection

Identification & Selection

First, a list of members who qualify for each award are presented to the team before utilizing a Google Form to anonymously vote for the award process.

The nominated members are sent to the Team Lead after a week to ensure that the nominated person meets the criteria (unless there is a conflict of interest). After confirming the nominee(s), for the Dean's List Award the nominee(s) are sent to the Head Mentor for confirmation.

Writing

After the Selection process for the Dean's List Award and Compass Award, the team is invited to participate in writing both the essay and script for the respective award.

4 people will be selected to write about the recipients and are given 2-3 weeks to complete their respective assignments, before they are sent to another member to peer reviewed.

Editing

After two members of the awards committee have fully written and reviewed each award, the essay/script is sent to the Team Lead and Historian to edit. They receive one week to fully review, modify, and change the essays where required while staying true to the original author's intent. Once approved by both the Team Lead and Historian, these are sent to the mentors for approval.

Finalization

Once the mentors receive the essays, they receive one week to review content. This is then submitted through the FIRST portal.

The secondary mentor is the sole reviewer of the Compass Award and will in turn suggest edits in order for finalization to take place before the Award Video is submitted.

Lessons Learned

Marketing

Rookie Team

As a rookie team, we started this year not knowing anything about marketing strategies. Due to this, we failed to market ourselves properly for several months before we discovered Social Media and researched other marketing strategies. Now, we have a list of ways that we plan to start marketing ourselves in the future to secure more sponsors and we are excited to do better!

Social Media

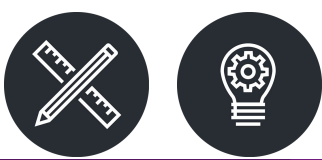
Impact

When we started posting on social media, we failed to realize the impact that social media actually has and so we lacked a consistent online image. However, we learned by posting more trendy videos that our videos do make it out there as we are able to secure both supporters and awareness of our team. This was an exciting lesson to learn, and we now know how to approach social media.

Awards Process

Time

We started the Awards Process too late this year, which taught us the importance of how much time each step of the awards process actually takes. This helped us secure when to start this process next year.



Design Process

Brainstorming

We began the design process by brainstorming and creating our own solutions for each part and assembly. Once we had several concepts, we researched existing RI30H designs to compare ideas and identify the most effective approach for our robot.

1

Prototypes

We created CAD models of three design concepts and evaluated them based on electrical integration, programming feasibility, available resources, and time constraints. This helped us select the most effective and buildable design.

2

Test solution (Lessons Learned)

Added tolerances to 3D-printed parts to prevent them from being too tight to fit together on the robot.

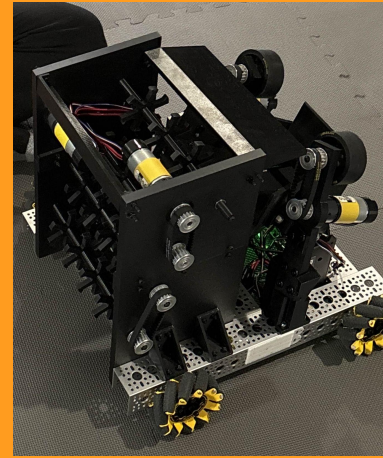
Testing the physical robot is critical.

Rushed CAD modeling caused design failure.

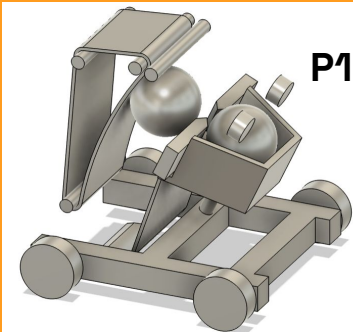
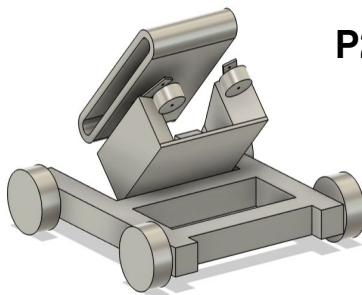
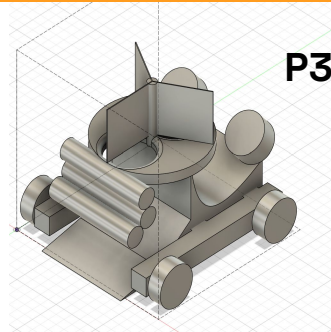
Switched to stronger shooter motors to achieve the needed velocity.

3

Final Solution

**4**

Prototypes

**P1****P2****P3**

Prototype #1 was selected as our robot design.

Reason Not Selected (P2): Could not support 3 artifacts at the same time and would take too long to tune comparatively

Reason Not Selected (P3): Robot designs must include space for an electrical board, and there was no way of actually shooting the artifacts.

CAD Tools

We found Fusion 360 to be an excellent CAD tool because it allowed us to design, simulate, and assemble our parts all in one platform. Its parametric features made it easy for us to adjust dimensions as our designs evolved. Fusion 360 also helped us collaborate effectively, track versions, and prepare parts for 3D printing and manufacturing. Its user-friendly interface and library of parts made it accessible and efficient for our team throughout the design process.

Our CAD Strategy

We approached our CAD process one part or assembly at a time, first outlining the necessary base components for each piece. We referenced existing parts when needed and ensured that each component was supported at multiple points for stability. To maintain consistency, we referenced hole locations on the frame and used standard measurements (4 mm, 8 mm, 16 mm) across the design. This methodical approach helped us create precise, well-integrated models that were easy to build and assemble. By following this strategy, we were able to quickly identify potential design issues before physical prototyping, saving time and resources.



Hardware

11/15



Rev Control Hub: Primary control system which powers the drive train motors, servo for the shooter, and handles main processing

Rev Expansion Hub: Used for additional motors for the intake and the motors on the shooter.

Rev Batteries: Main power sources mounted between the control hub and expansion hub using a strap, we own 5 for better outcomes during competition.

6 GoBILDA 321 RPM motors: Used for drivetrain and intake. (With the addition of 2 GoBILDA 6000 RPM motors for shooter)

4-Pos JST PH to 4-Pos JST XH Adaptor (600mm Length): Provides data to the robot

2-Pos JST VH to 3.5mm Bullet Adaptor (100mm Length): Provides power to the motors.

2 GoBILDA Odometry sensors: Provides continuous x and y displacement and is used for field tracking.

1 GoBILDA Servo: Used to push artifacts into the shooter.

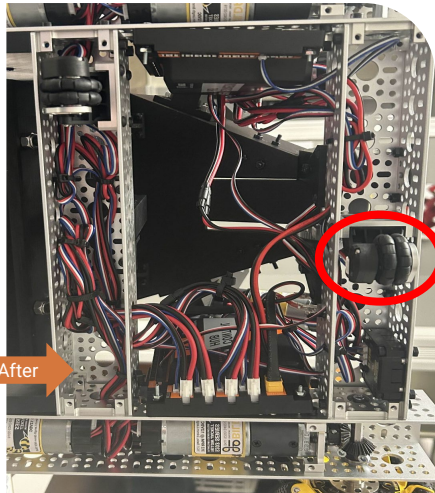
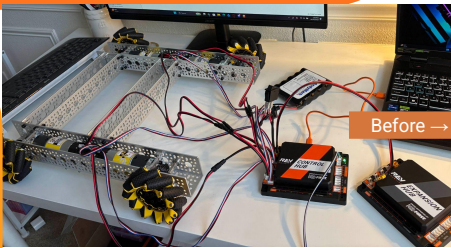


Wiring Standards and Cable Management

Cable routing was intentionally planned to protect wiring and improve long-term reliability. All cables were routed along designated paths beneath the chassis, keeping them clear of moving mechanisms and potential pinch points. Motor power and encoder signal lines were grouped by subsystem and securely fastened to minimize strain, vibration, and accidental disconnections during match play. The final wiring layout was designed to remain clean, accessible, and easy to manage. This organization allowed quick inspection and maintenance without removing major components.

PinPoint Odometry

We implemented a dual-wheel odometry system to track the robot's position on the field accurately. It tracks the x and y coordinates of the robot and provides the data to us for a more smooth and optimal autonomous performance. It helps with a more consistent autonomous period and corrects any errors. Odometry encoders are wired through JST-PH to the Control Hub and the cables are routed away from the drivetrain (Wiring is circled on the "After" picture).



Reliability

Stress tests of motors and servos were conducted for functionality and for purposes of battery utilization. **Battery Health** checks are conducted before and after every match to ensure reliability. There were few failures throughout competition, and those that appeared had simple fixes.

Electrical Wiring Standards

To ensure consistency and reliability across all robot subsystems, we established standardized wiring practices. XT30 cables were used for all motor power connections to provide secure, high-current delivery, while JST-PH encoder cables were used for all data and feedback signals. Standardizing connectors across the robot reduced wiring errors, simplified installation, and made component replacement more efficient. Each motor, encoder, and servo connection followed the same wiring conventions, allowing team members to quickly identify correct ports and verify configurations during inspections and testing. All wires and motors were clearly labeled to support troubleshooting.

Lesson Learned: Early in the build process, motors were not labeled, which resulted in configuration errors and required rewiring. This experience reinforced the importance of labeling both components and wires during initial installation to reduce debugging time and improve system reliability.

Control System Overview

Our robot uses closed-loop control and sensor feedback to achieve reliable autonomous performance and consistent driver control. Odometry and motor encoders provide real-time data that allows the robot to correct errors, maintain repeatability, and perform predictably under match conditions. To support our eight-motor configuration, we used a REV Control Hub and Expansion Hub. The drivetrain was powered by the Control Hub, while the intake and shooter were powered by the Expansion Hub, reducing electrical load concentration and improving system reliability. A single servo is used to transfer Artifacts.

Potential Improvements

Protect the expansion hub and control hub, although they are safe, they are screwed in standing up on the sides of the robot. Make battery mounting better and faster, while we unmount or mount the battery, it takes a little time and it could potentially unplug the power cable for the control hub so it makes the person remove it gently which takes up a little time.

Maintain a log for each electrical modification in the future, so we can get our parts earlier and also keep a consistent record.

Design Process

Mapping & Planning	Prototype	Testing	Finalization
We determined the components we wanted based on the base control system and what we would need to accommodate the subsystems. Based on the robot CAD, we planned out positioning and wiring.	Once we received the robot, we attached wires based on our prior plan and began the process of wire management.	We then tested each component of our electrical system to ensure that it would function properly throughout competition. We found that the wiring wouldn't make sense to anyone who needed to fix something fast.	We improved our wire management, by creating clean and simple wiring beneath the chassis of the robot. We also labeled every wire and component.



Software

PID

We used PID and FeedForward control on the shooter mechanism and the drivetrain as well. In doing so, we were able to maintain accuracy during the match.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Software Development & Teaching

For accessibility, most programming meetings were held virtually and code is required to have comments to ensure people with less experience were able to learn and therefore develop robot code.

Additionally, documentation/ training materials are provided to all members.

Reliability TeleOp

TeleOperation code has proven to be consistently reliable throughout both practice and was consistently reliable throughout our first qualifying competition.

Reliability Autonomous

We faced consistent issues within our autonomous period during our first competition due to a localization error. After taking more accurate measurements and testing, we were able to fix it.

Programming Process

Define the Problem

In order to complete this part of the process we focused on two problems:

- Hardware
- What we need to accomplish

In order to break this down further, the robot was broken into subsystems within the code:

- Shooter:
 - Two Motors + One Servo
- Intake
 - Two Motors
- DriveTrain:
 - Four Motors
 - Pinpoint + Deadwheels
 - Internal IMU

Development

Design software infrastructure:

- Command-Based Programming
 - Define classes
 - Define basic commands
 - Use basic commands to build complex ones
- Define how subsystems will interact with each other
 - (e.g. intake feeds into shooter, so timing between their actions matters)

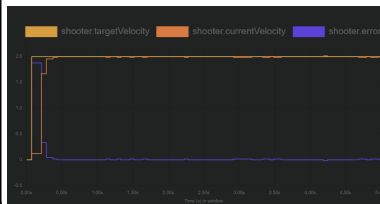
Testing

Test code to see performance:

- Use initial PID values to see performance
- See initial path performance

Refine Robot:

- Tune PIDF values
- Modify paths



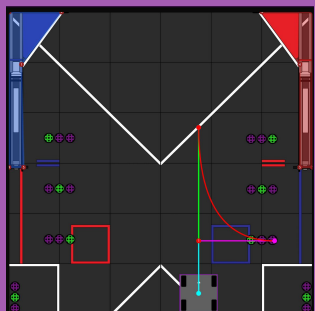
Final Product

```
//First begin with a command defined earlier
Command servoAlternate = servoNeutralPosition() Command
//If goal within preset tolerance, continue
.then(new WaitUntil()->Math.abs(
    controlSystem.getGoal().getVelocity()
    -shooterMotor.getVelocity())
    < Constants.TOLERANCE)): SequentialGroup
//Use a command defined earlier
.then(servoShootPosition())
//Wait for game piece to exit
.then(new Delay(Constants.SHOOT_DELAY))
//Clean up/reset mechanism
.then(servoNeutralPosition());
```

PedroPathing

We used PedroPathing as our path planning library, incorporating data from two deadwheels and the GoBilda Pinpoint computer/its IMU. As a result, we were able to have accurate autonomous pathing and control our robot with a great degree of precision.

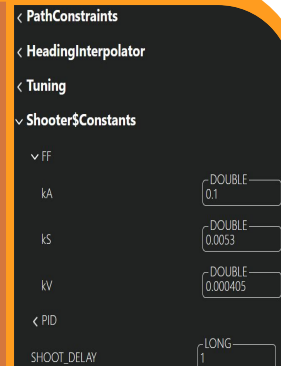
Localization: For localization, we made use of two deadwheels alongside a GoBilda Pinpoint computer, to provide accuracy during the autonomous and teleoperated periods. We are using PedroPathing, for both its beginner friendliness and efficiency.



Live Updating

Redeploying code every time a value needed adjustment would have consumed a significant amount of time, so our team implemented the Panels library. This greatly reduced tuning time, allowing us to spend more time practicing with and further developing our robot.

The image to the right is one of the tuning dashboard, which allowed us to efficiently iterate on our mechanisms.



Potential Improvements

- Incorporate vision
- Cut down delays in code to minimum working levels to reduce time wasted in match
- Add indicators to give drive team more info (e.g. LEDs to see robot status)
- Add dynamic shooter velocity

Developer Tooling

Java using Android Studio. All code is maintained on GitHub

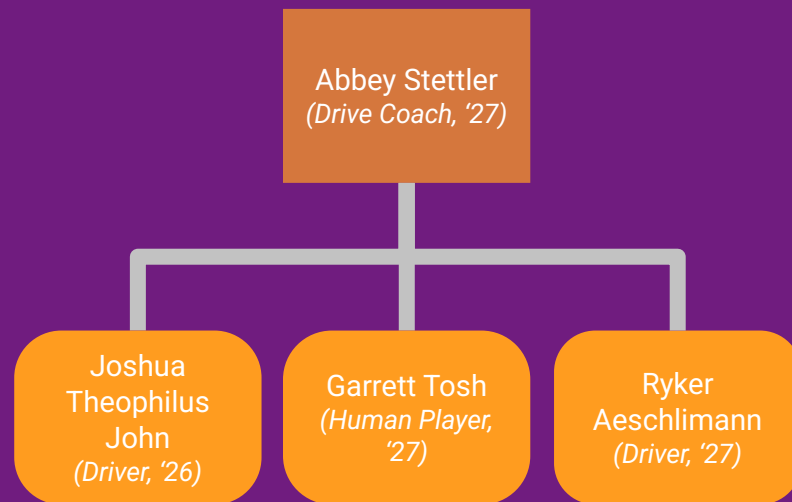
In addition to the FTC SDK, we utilized the NextFTC and Pedro Pathing libraries to enhance functionality and development efficiency.



Drive Team

Drive Team Selection

Our drive team is a small group of students selected for their teamwork, consistency, and positive attitude. They work closely together and represent our team during matches and events. This group is trusted to perform under pressure while exemplifying Gracious Professionalism and upholding the core values of FIRST. Team members were selected through an application process that was reviewed by the Team Lead and Head Mentor to ensure the best possible candidates were chosen to exemplify the role and its responsibilities.



Alliance Coordination

Our drive team works closely with alliance partners to develop effective match strategies. Before each match, we communicate to define roles based on each robot's capabilities and select the human player best suited for the alliance strategy. By focusing on consistent contributions within our strengths, we help support and execute the overall alliance plan.

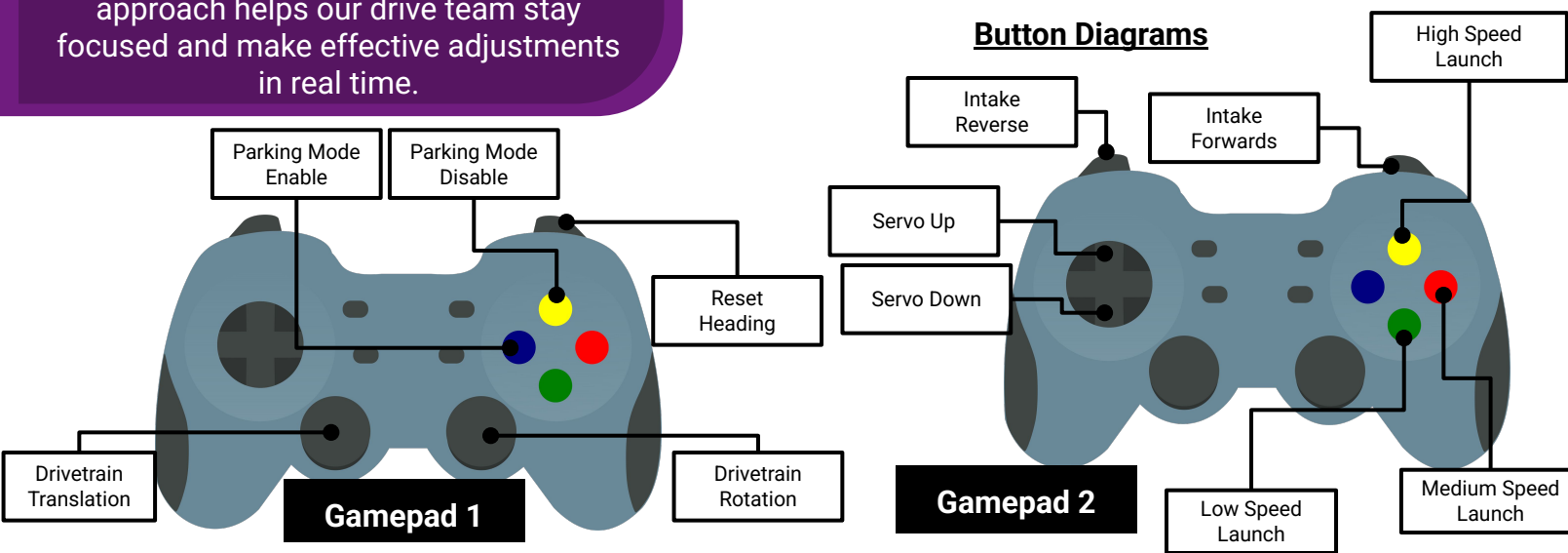
Communication & Adaptability

Clear and respectful communication is essential to our team's success. During matches, we prioritize calm, concise callouts that enable us to adapt quickly to changes in the field or robot performance. We foster a positive environment, avoiding drama or bullying, to ensure all communication remains productive and respectful. This approach helps our drive team stay focused and make effective adjustments in real time.

Match Strategy

Because our robot is unable to shoot Artifacts in a specific order, our drive team focuses on manually collecting and scoring Artifacts toward the end of TeleOp. When possible, our alliance partners prioritize ordered scoring, while we concentrate on consistent pickup and reliable scoring. This strategy enables our team to contribute meaningful points despite our robot's limitations.

Button Diagrams





Season Planning

Summer (Pre)season: Planning for the upcoming seasons, Member registration, Leadership selection, Subteam education.

May - August

Practice Season: Drive team selection, Strategy team selection, Driver/Human Player practice.

November - January

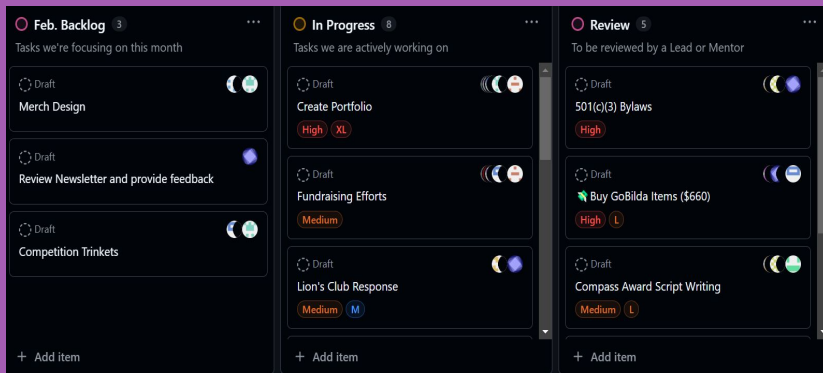
September - November

Build Season: Kickoff, Brainstorming, CADing, Manufacturing, Programming, Testing.

January - April

Competition Season: Qualifying Competitions, FIN State Championship, FIRST World Championship

Team Objectives & Tracking



We track our top priorities with a task board. Each meeting to keep every subteam aligned and moving forward

Base Budget Planning

Cost	Item	Supplier
300	Team Registratic	FIRST
325	Event Registratic	FIN
275	States Event Re	FIN
1025	Full FTC Field	AndyMark
485	DECODE™ Piec	AndyMark
285	Driver Kit	KoP (REV?)
325	Electronics Kit	KoP (REV?)
250	Expansion Hub	REV
165	Robot Battery (x	REV
80	Battery Charger	REV
900	Robits Core Kit	AndyMark
210	Odometry Pack	GoBilda
600	Event Food (x3)	
375	Shirts (x25)	
200	Insurance	
450	Buffer / Misc	
\$6,250 < Total		

We planned our base budget beforehand so we can prepare and outline the essential investments required to launch and sustain our season

Seasons Costs

Category	Sum of Amount
Apparel	\$437.00
Field	\$1,582.95
Food(Estimated)	\$600.00
Registration	\$625.00
Robot	\$2,636.11
Grand Total	\$5,881.06

Our season costs outline our core investments needed to build, compete, and fuel our team throughout the season

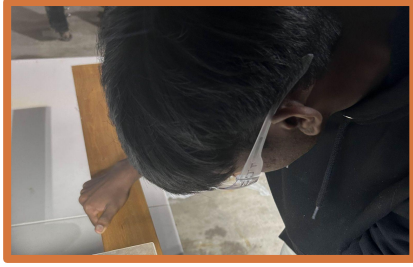
Funds

Category	Sum of Amount
Apparel	\$107.00
Donations	\$1,755.63
Grants	\$2025.10
Student Dues	\$4,440.00
Grand Total	\$8,327.73

Our funds outline our sources of income within the team which allow us to pay for all of our season's costs.



Reflections



"I liked that we came together as a team and was able to participate in the competition and show **amazing teamwork with a very diverse set of students**, and I also believe that we have learned so much and each member is also involved in the progress which expands the focus to every subteam and provides everyone with immense knowledge which has also benefited me in my own life. I believe that we can work on scoring a little more frequently and consistently, and get used to the robot a bit more and have more time with it."

← *Joshua Theophilus John, Integration Director & Driver*

"I have gained a lot of knowledge on FTC in general. Before this past summer, I knew hardly anything about FTC. I never even when up to the FTC fields at Worlds because I was busy. It has been **really cool to see just how far I've come this year.**"

Abbey Stettler, Mechanical Primary & Drive Coach →

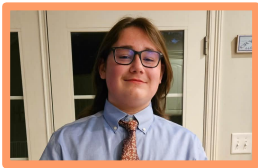


"I feel like we've all done a good job on drive team, I think the only issue has been communication, I'm a part of that problem and I'm working on it, but after that all gets worked out and we figure out weeks and such, **I believe we'll do great at comps.**"

← *Ryker Aeschlimann, Business Primary & Driver*

"For never having done anything with robotics before, I am surprised but happy with the learning experience I've had. **The community inspires me to always work to do my best.**"

Caroline Kerr, Historian & Awards Director →



"The robot worked very well for the resources and budget provided. **I am very proud of how effective it is** and can't wait to see it hopefully get better next year."

← *Nathan Couch, Mechanical Primary & Business Secondary*

"As the founder of this team, I am incredibly proud of how far we have come. We were literally built from scratch and in the garage's of some of my favorite people in the world. **I'm beyond proud of all of our progress**, from building a functioning robot to winning the Connect Award and I'm so excited about our future."

Sage Ryker, Team Lead & Founder of Orion | NorthStarRobotics →



DECODE Season Reflections

COMMUNITY

Throughout this season, our team has forged meaningful connections with several organizations which we are excited to continue nurturing. Equally important, we've cultivated a strong sense of community within our own team, creating friendships that will undoubtedly endure long after this season ends.

ROBOT

This year our build season faced challenges due to a lack of structure, resulting in delays that made it difficult to stay on track. The absence of a set timeline left us feeling disorganized and slowed our progress. However, we have learned valuable lessons from this, recognizing the importance of better planning and clearly communicating moving forward.

FUNDRAISING

This year, we had some success with community sponsorships, which played a key role in our fundraising efforts. However, we relied heavily on student-paid dues to meet our financial goals. This raised concerns about the sustainability of this model, especially where dues to increase in the future. Moving forward we recognize the need to begin our fundraising efforts earlier.

COMPETITION

Our goal for this competition season is to ignite a larger passion within every team member. While winning is always a great achievement, our true focus is on building a love for robotics and STEAM throughout all of our students. We aim to create an environment where the excitement of learning and growth takes center stage, encouraging everyone to develop both their skills.